

Yousef Ehab Disoki

Communication and computer Engineering Student

yousef.disoki.5127@gmail.com | +20 1280666843 | Shoubra, Cairo, Egypt | [linkedin](#) | [github](#)

SUMMARY

Communications Engineering student with a strong interest in railway systems, electrical safety, and site engineering. Gained practical exposure to railway signaling, SCADA, technical documentation, AutoCAD, requirements management, and Verification & Validation through training at Alstom. I enjoy applying engineering concepts through hands-on projects, including a Flutter-based LOTO safety simulator focused on equipment isolation and maintenance safety. Motivated to learn, work in dynamic site environments, and contribute to large-scale transportation projects.

EDUCATION

Bachelor of Engineering in Computer and Communications Engineering

Shoubra Faculty of Engineering, Benha University

Cairo, Egypt

Sep 2023 – Jun 2028 (Expected)

- **Relevant Coursework:** Electric Circuits, Electronics, Electromagnetic Fields, Electrical Measurements, Control Systems, Digital Signal Processing

TECHNICAL EXPERIENCE

Railway Signaling & Systems Engineering Trainee

Alstom

5th settlement, Cairo

Jul 14, 2024 – Aug 5, 2024

- Studied requirements definition and configuration management principles for railway signaling systems, gaining exposure to requirements traceability and change control practices used in safety-critical rail engineering
- Trained on core signaling systems including Interlocking Systems (IXL), Integrated Control Centers (ICC), Automatic Train Supervision (ATS), and Automatic Train Control (ATC), developing applied knowledge of safe train routing, and automated train operation principles used in live rail networks
- Completed an introduction to Traffic Management System (TMS) IT architecture and SCADA-based automation platforms used for real-time monitoring and control of railway operations across multiple stations
- Completed an introduction to cybersecurity principles for protecting critical railway control infrastructure against operational and digital threats
- Studied Verification & Validation (V&V) processes for safety-critical railway systems, developing awareness of compliance requirements under international railway safety standards
- Produced technical documentation and system layout drawings in AutoCAD as part of railway infrastructure planning exercises

TRAININGS & CERTIFICATIONS

Machine Learning for Data Analysis (120 hrs)

Creativa Innovation Hub, NTI

Nasr city, Cairo

Apr 15, 2026 – Jul 27, 2026

- Completed 120-hour program covering the full ML pipeline: data preprocessing, feature engineering, and exploratory data analysis (EDA) through supervised learning (linear/logistic regression, decision trees, random forests, gradient boosting, SVM, neural networks) and unsupervised learning (k-means, PCA)
- Applied model evaluation techniques including cross-validation, hyperparameter tuning, and performance metrics using scikit-learn and TensorFlow on real-world datasets
- Completed a freelance career track covering portfolio building, client proposals, and pricing strategy for delivering data analysis services independently

PROJECTS

Lockout-Tagout (LOTO) Safety Simulation App

Flutter, Dart

[github](#)

Jun 2026

- Developed an interactive application simulating Lockout-Tagout (LOTO) procedures for safe equipment isolation before maintenance activities.
- Implemented a structured workflow for identifying hazardous energy sources, equipment shutdown, isolation, lock/tag application, verification, and controlled re-energization.
- Designed safety checks and procedural guidance to demonstrate principles of electrical isolation, maintenance safety, and EHS compliance in industrial environments.

Flight Route Planner

Java, JavaFX, Graph Algorithms, Dijkstra

[github](#)
May 2026

- Built an interactive flight route optimization tool using a weighted directed graph, implementing Dijkstra, A*, and BFS algorithms and comparing their performance across 50-node and 200-node test graphs
- Reduced A* average path computation time by 38% over Dijkstra on sparse graphs through an admissible heuristic based on haversine distance estimation

VOLUNTEERING

Team Liaison Officer (TLO) for Turkey and England teams

Para Badminton International 2026

New Capital Sports Hall, Egypt
Jan 13, 2026 – Jan 18, 2026

- Delivered liaison support and accessible logistics coordination for Turkey and England para-badminton delegations across competition and training sessions

Hall Director

IHF Mens Super Globe 2025

New Capital Sports Hall, Egypt
Sep 25, 2025 – Oct 2, 2025

- Managed full hall operations for an IHF top-tier club tournament, overseeing staff coordination, crowd flow, and real-time compliance with IHF event standards

TECHNICAL SKILLS

Railway & Control Systems:	SCADA, IXL, ICC, ATS, ATC, AutoCAD
Electrical Safety:	LOTO Principles, Hazardous Energy Isolation Awareness, Electrical Safety Awareness
Documentation:	Technical Documentation, Configuration Management, V&V
Microsoft Office & Data:	Excel, PowerPoint, Word, Data Analysis, Technical Reporting
Programming:	Python, Dart, Java, Flutter

LANGUAGES

- **Arabic** – Native / Full Professional Proficiency
- **English** – Professional Working Proficiency (B2 level)
- **German** – Beginner (A1 level)